23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – 2 (Design)
Architectural Modelling (CO2)

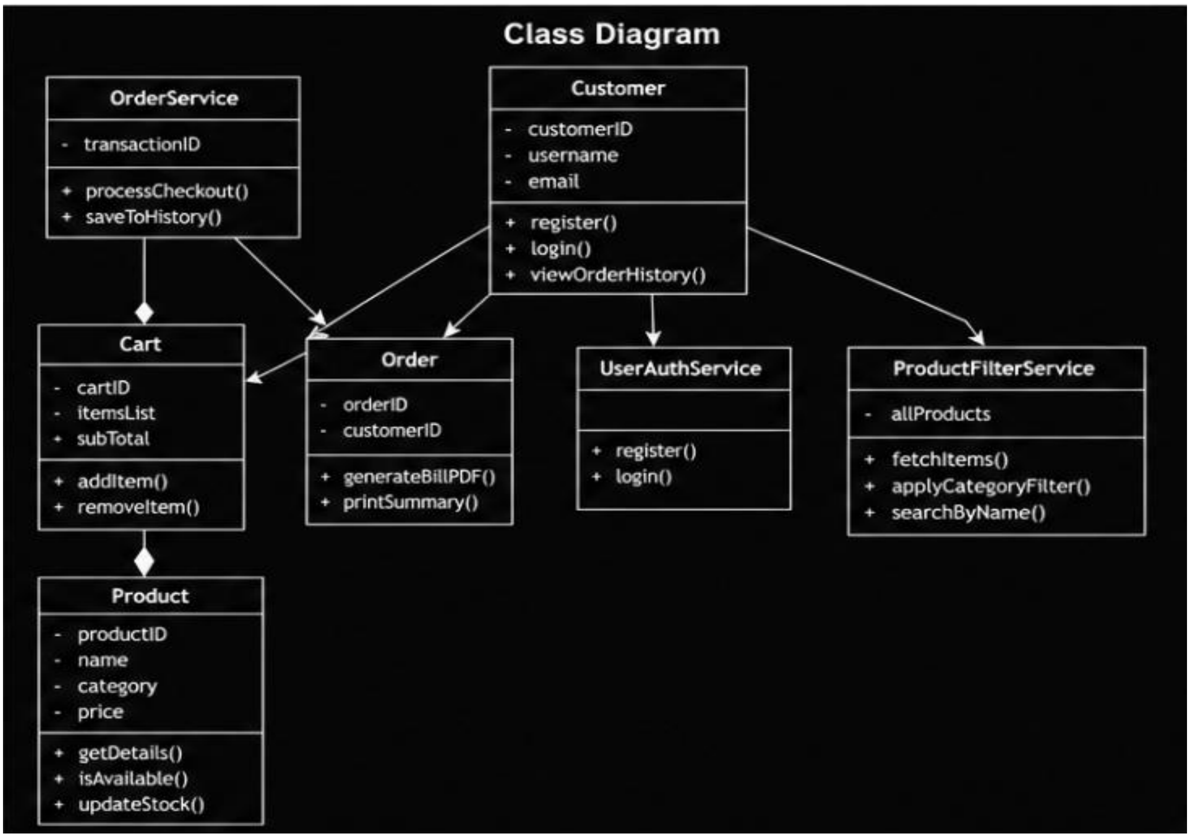
Group No: D6
Team Members :

Roll No	Student Name
AM.SC.U4CSE25346	KARTHIK SANKISA
AM.SC.U4CSE25333	M. PURI JAGANNADH
AM.SC.U4CSE25309	B.SAI ASHISH
AM.SC.U4CSE25343	R.VEDANANDA REDDY

1. Problem Title : ONLINE SHOPPING SYSTEM

STRUCTURAL DESIGN

2. Class diagram (*draw class diagram that shows different classes and their relationships. Use UML*



BEHAVIUORAL DESIGN

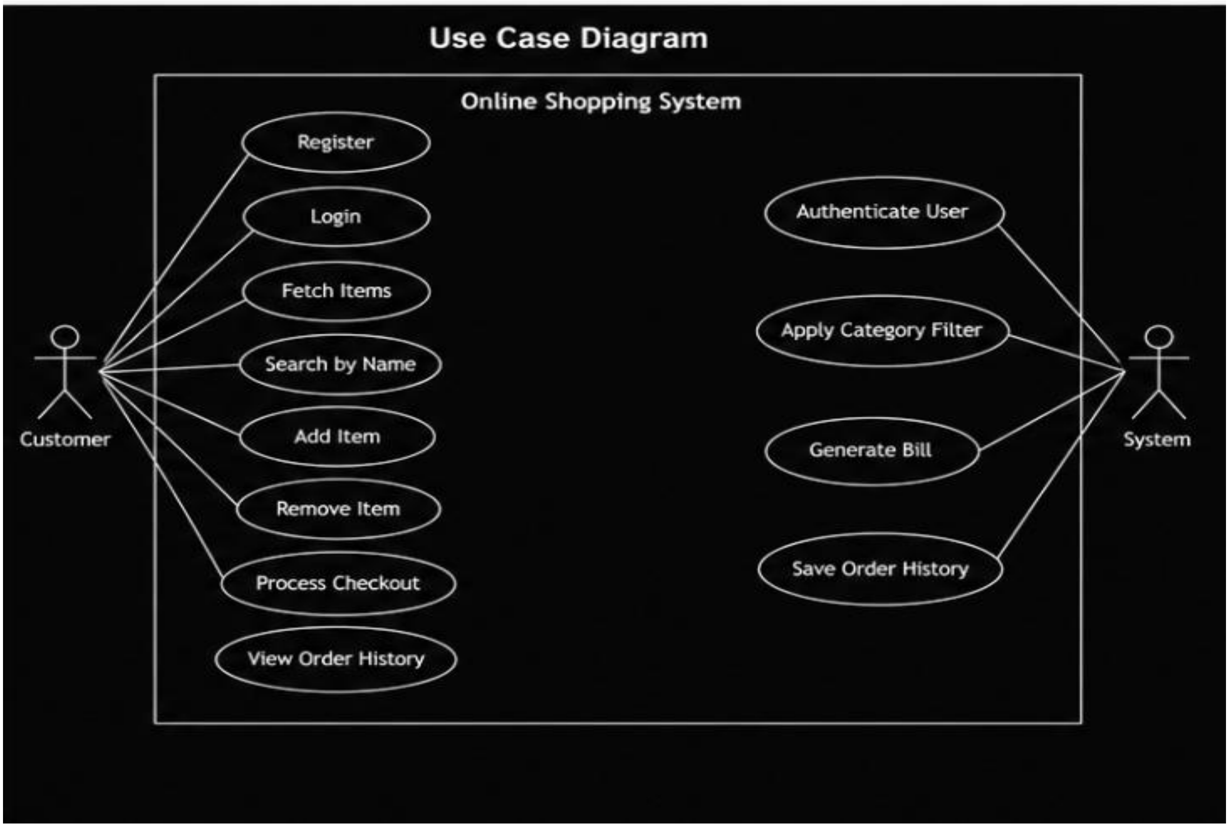
a. List of Actors

1. Customer
2. System (Services)

b. List of Use Cases

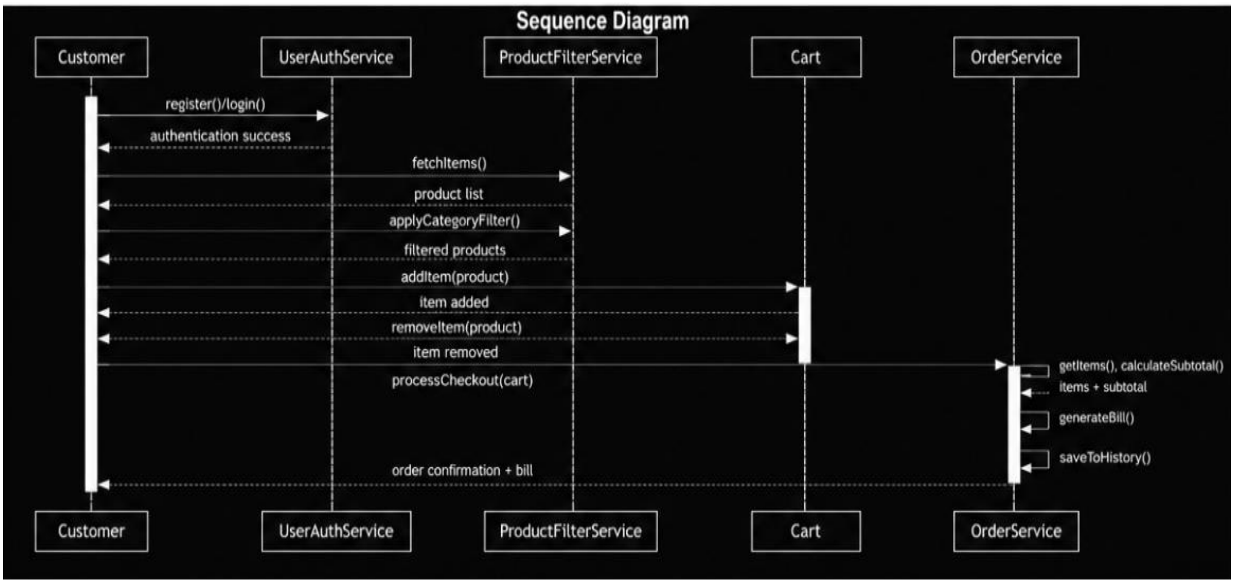
1. Register/Login
   Customer creates an account or logs into the system.
2. Browse Products
   Customer views available products and applies filters.
3. Add to Cart
   Customer adds selected items to the cart.
4. Remove from Cart
   Customer removes items from the cart.
5. Checkout
   Customer places an order and completes purchase.
6. Generate Bill
   System generates PDF bill after order placement.
7. View Order History
   Customer views past orders.Use case diagram   (use UML notations)

c. Use case diagram(UML Notations)



4.Sequence diagram (use UML notations)

a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.



Relationships Used (Association / Aggregation / Inheritance)

Inheritance:

There is no inheritance hierarchy explicitly defined in this system. All classes operate independently with specific responsibilities.

Association:

- Customer is associated with Cart, Product, and Order.
- Cart is associated with Product (a cart contains selected products).
- OrderService is associated with Cart and Order for processing transactions.
- ProductFilterService is associated with Product for searching and filtering.

Aggregation:

- Customer aggregates Cart and Order, as one customer can have multiple carts/orders over time.

Composition:

- Cart composes Product items, since the cart directly contains selected products.
- Order composes the final transaction details, including selected items and billing information.

Key Interaction Flow Captured in the Sequence Diagram

The system starts with the customer interacting with the application.

1. The **Customer** registers using register(customerID, username, email) and logs in using login(username, password) through the **UserAuthService**.
2. After successful authentication, the customer browses products using fetchItems() and applies filters using applyCategoryFilter() or searchByName() via the **ProductFilterService**.
3. The customer selects products and adds them to the cart using addItem(product) and may remove items using removeItem(product) in the **Cart**.
4. When ready, the customer proceeds to checkout using processCheckout(cart) handled by the **OrderService**.

5. The **OrderService** retrieves cart details, calculates the total, and generates a bill using generateBillPDF().
6. The order details are stored using saveToHistory() for future reference.
7. Finally, the system confirms the successful order placement and allows the customer to view previous orders using viewOrderHistory().

Major Classes Identified and Their Responsibilities

Customer

Represents the user of the system.
Stores attributes such as customerID, username, and email.
Responsible for registering, logging in, and viewing order history.

Product

Represents items available in the store.
Contains attributes like productID, name, category, and price.
Responsible for providing product details, checking availability, and updating stock.

Cart

Represents a temporary container for selected products.
Stores cartID, itemsList, and subTotal.
Handles adding and removing items and calculating the total cost.

Order

Represents a finalized transaction.
Contains orderID and customerID.
Responsible for generating the bill and summarizing the order.

UserAuthService

Handles authentication operations.
Responsible for user registration and login validation.

ProductFilterService

Handles product searching and filtering.
Responsible for fetching product lists, applying category filters, and searching by name.

OrderService

Manages order processing.
Handles checkout operations, bill generation, and saving order history.